Synthesizing Some Australian Birds

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Abstract

This paper describes an additive synthesis design for synthesizing bird calls which models their dynamic spectra and many of their expressive characteristics. The Csound design produces tones that are very similar to several Australian bird calls. The design also works for other animal sounds, such as crickets. Finally, excerpts from a musical composition illustrate the musical uses of the design and the flexibility of varying the calls.

Introduction

When hiking around Australia in 2007, we saw some beautiful birds (see Figure 1) and listened to some attractive bird calls. Although it is a mystery whether the silent birds we saw made any of the calls we actually heard when we weren't looking at them, we've synthesized some of the calls using additive synthesis in Csound (Vercoe 1992).



Figure 1. Birds at Uluru

This paper first gives some background on sound synthesis research and then describes our source material. As the additive synthesis design is very similar to a previous design, we focus mainly on the spectral data used for synthesizing bird calls. We then discuss several examples of the bird call transcriptions. Finally, we conclude with excerpts from a composition illustrating the musical uses of the design with the flexibility of varying the calls, and an extension to synthesize a cricket.

Background

Ayers wanted to use some sampled bird recordings in a composition, but they contained a lot of background noise which filtering did not sufficiently remove. Could our previous synthesis design for musical instrument tones model the bird calls? She tried it without initially considering the research implications of the project. The goal was simply to replace the recordings, and the method worked. On further reflection, when we realized that most of the bird calls used in computer music compositions that we've heard have been sampled, and that few have been synthesized, we decided to share our results with the community.

We previously designed group additive synthesis models of many wind instrument tones (Horner, Ayers and Law 1999; Horner and Ayers 1998a, 2002) and synthesized timbre tremolos and flutter tonguing on wind instruments (Ayers 2003, 2004; Ayers and Horner 2004a and b). We have synthesized the Chinese flutes and the *suona* using additive synthesis (Ayers 2005; Ayers and Horner 2006), and we have also synthesized the *didgeridoo* using timbre morphing (Ayers and Horner 2007).

Other researchers have focused their attention on the acoustics (Greenewalt 1968; Fletcher and Tarnopolsky 1999) or physical modelling of bird calls (Kahrs and Avanzini 2001). Some of these models are quite complex and their goal seems to be pure research rather than creating a bird instrument to use in compositions. Our research is basically working in the opposite direction: while most of the other researchers are trying to synthetically reconstruct the bird and consider how its physical characteristics affect the sound, we're trying to reconstruct the sound without considering the shape of the bird that made it. We found one additive synthesis application in Matlab, "The Additive Synthesis Forest," which doesn't seem intended as a composition tool (Sturm 2003)..

The Source Material

We recorded some attractive bird calls when hiking around Watarrka, Uluru, Kata Tjuta and Alice

Springs in central Australia, Namadji National Park near Canberra, and around Cairns in 2007. We recorded the sounds using the only recording device we had brought with us, our digital video camera. With the environmental background noise and camcorder quality, it was less than optimal, but we got very interesting synthesis results in spite of the limitations. Our motivation for synthesizing the sounds was to clean up the background noise which made them difficult to use in a sampled instrument design.

In order to make the recording usable, we cut them up into the cleanest segments, and then filtered them to mimimize the background noises, such as wind, camera clicks, other birds, etc. Although such a process is necessarily "lossy," the soundfiles we used produced a pretty fair representation of the bird sounds. Figure 2 shows a spectrogram (Horne 2001) analysis of a short segment used for the synthesis, with four calls from one bird, and two calls from another bird in the background. The pitch contour is fairly flat for the fundamental of the bird call we want, at around 2000 Hertz, and the second bird is lower, around 750 Hertz.

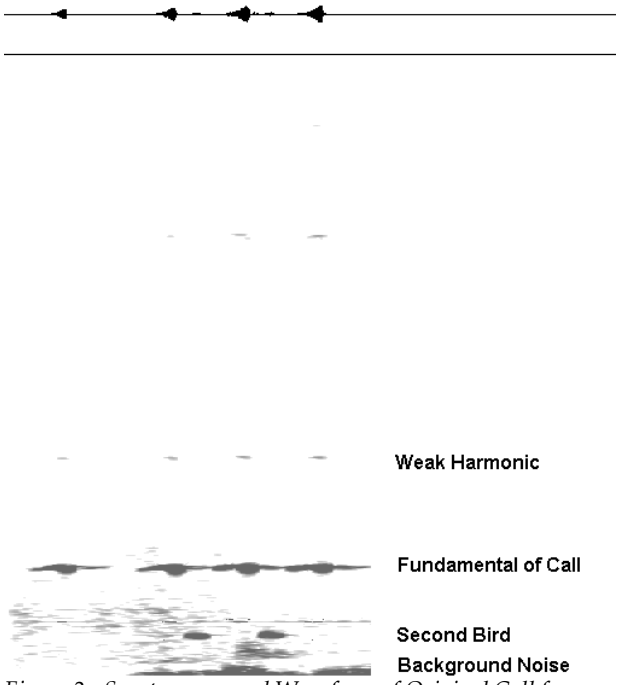


Figure 2. Spectrogram and Waveform of Original Call for Bird 1

In order to remove the hissy background and the two calls from the second bird, we filtered the bird calls. This produced the cleaner sound shown in Figure 3, with a further decrease in amplitude of the already weak higher harmonics.

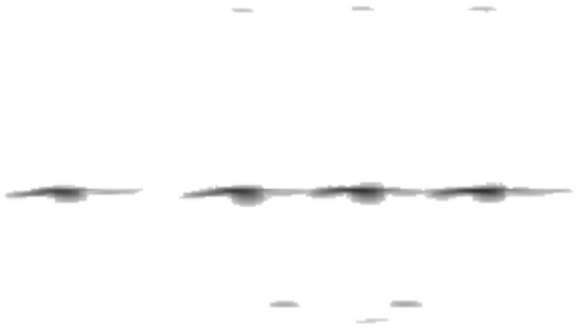


Figure 3. Spectrogram of a Filtered Call for Bird 1

The Additive Synthesis Design

The bird calls modelled in this paper are clearly pitched, but they don't divide up into 12-tone equal temperament scales. After examining the recordings, we found that we could get the musical results we wanted from synthesizing the average spectra of just three bird calls. We grouped two calls together for Bird 1, and kept one call separate for Bird 2.

Our previous additive synthesis designs use up to 63 sine waves to model the rich spectra of various timbres such as the *suona* (Ayers and Horner 2006). Most bird calls are higher than *suona* tones, their spectra aren't as rich in harmonics, and 10-20 harmonic sine waves were enough to model their timbres. Since the bird spectra worked well in the additive synthesis design, we won't review its code here. Instead, we will focus on how we customized the data to produce bird calls. An important feature of our design is that repeated performances of the bird calls will have varied spectra in order to make the performance sound more natural, which is useful for producing a more expressive composition.

Special Amplitude Envelope for "Bird 1"

As shown in Figure 2, the higher call for Bird 1 has an unusual amplitude envelope that rises for most of the sound and then decays very quickly. In order to handle this special case, we added a function table that produces the envelope with a slight randomized variation (see Figure 4). The minimum amplitude is .1% of the total amplitude and the maximum amplitude is 100% at about 85% of the duration of the tone. If this special envelope is not needed, the design uses a simple rise, sustain and decay.

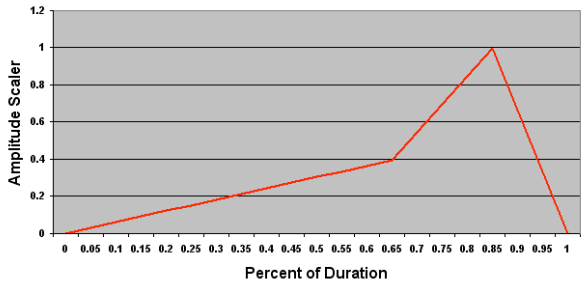


Figure 4. Amplitude Function for Bird 1

The Bird Spectra

This section describes the spectra of two groups of calls with similar spectra which we combined as Bird 1 and the separate call for Bird 2.

The Spectral Components for "Bird 1"

Figure 5 shows an 11-harmonic time-varying phase vocoder analysis (Dolson 1986; Wun 2000) of the group of four calls from one bird. The pitch contour is fairly flat, at around 2000 Hertz, and the segment contains mainly one loud low harmonic and much instability among the higher harmonics.

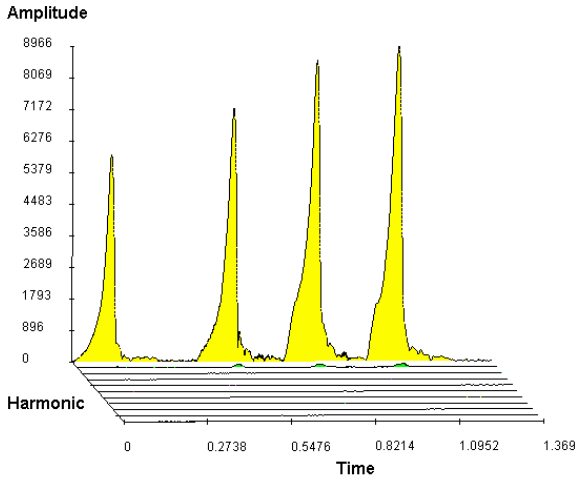


Figure 5. Time-Varying Phase Vocoder Analysis of Filtered 2000 Hertz Call for Bird 1.

We use the average harmonic amplitudes to model this bird call based on our previous additive synthesis wind instrument designs (see Figure 6). The first harmonic is quite strong and the others very weak. Based on this spectrum alone, we might consider just using a sine wave to synthesize the calls, as others have done (Assman). However, we found that the low amplitude harmonics above the fundamental do contribute to the bird call's timbre, particularly in the lower frequency calls described in the following discussion.

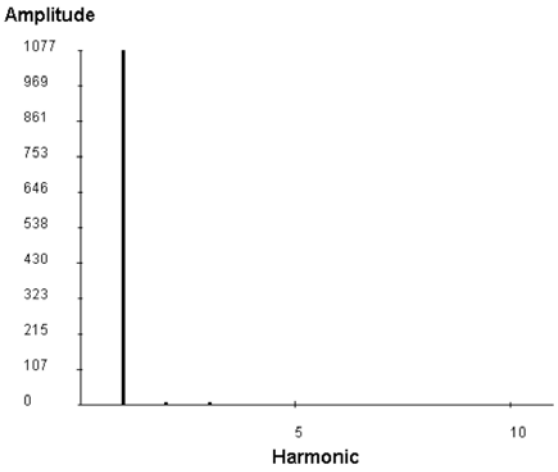


Figure 6. Average Harmonic Amplitudes of Filtered 2000 Hertz Call for Bird 1.

Figure 7 shows a 20-harmonic time-varying phase vocoder analysis of a lower bird call at 1100 Hertz. As with the higher call, the lower call contains mainly one loud low harmonic and much instability among the higher harmonics.

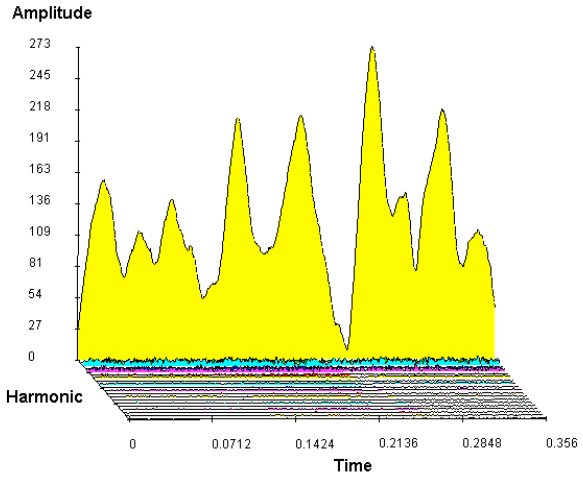


Figure 7. Time-Varying Phase Vocoder Analysis of Filtered 1100 Hertz Call for Bird 1.

Figure 8 shows the average harmonic amplitudes used to model this bird call. The first harmonic is still stronger than the others, but proportionally the amplitude of the other harmonics is stronger, with more noise in the higher harmonics. As the amount of variation in the higher harmonics is within the range of random variation incorporated in the design, we did not include additional function tables to cover bird calls between these two frequencies.

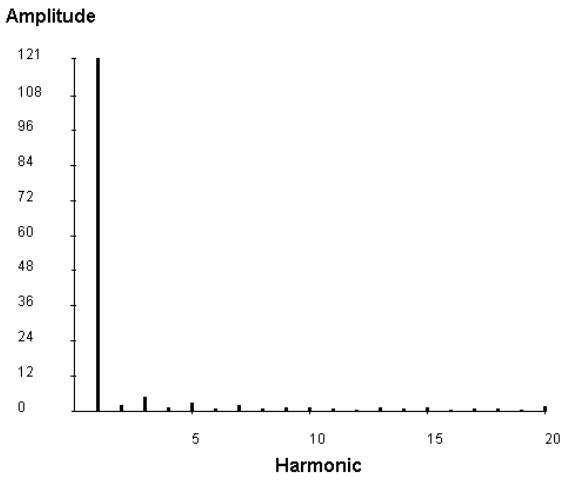


Figure 8. Average Harmonic Amplitudes of Filtered 1100 Hertz Call for Bird 1.

We stored the amplitude data from the graphs in two Csound score function tables (see Figure 9).

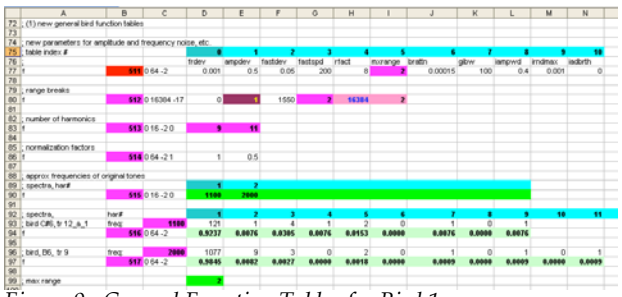


Figure 9. Csound Function Tables for Bird 1

The Spectral Components for "Bird 2"

Although most of the calls have spectra similar to the average amplitudes used for Bird 1, a few calls have a much stronger second harmonic and a very weak fundamental. Did we make an error in finding the fundamental frequency of these calls? As the synthesis for these calls really does work better with the weak fundamental, strong second harmonic spectrum than it works with lowering the fundamental by an octave and using either of the other spectra, we separated it from the other tones as "Bird 2."

Figure 10 shows a spectrogram analysis of the call used for Bird 2, with six similar calls, each preceded with a portamento grace note. The fundamental frequency of this call is around A6 at 1820 Hertz, with the grace note beginning at around 1650 Hertz (about G#6).



Figure 10. Spectrogram of Filtered Call for Bird 2

Figure 11 shows a 12-harmonic time-varying phase vocoder analysis of the segment. The segment contains mainly one loud second harmonic and much instability among the higher harmonics.

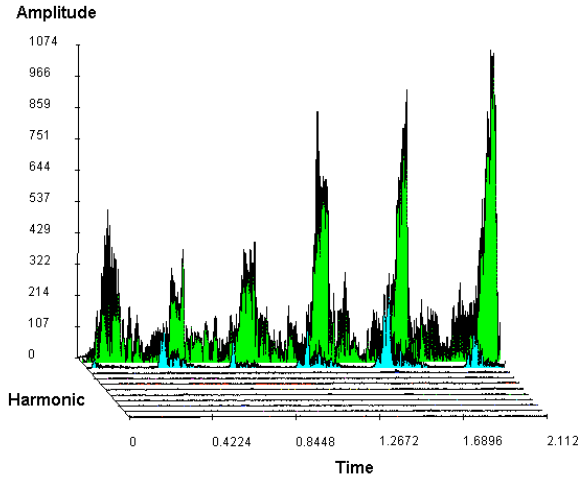


Figure 11. Time-Varying Phase Vocoder Analysis of Six Filtered 1820 Hertz Bird Calls.

Figure 12 shows the average harmonic amplitudes used to model this group of calls. The second harmonic is stronger than the others, with harmonics 1 and 3 also having significant amplitudes and little amplitude in the higher harmonics.

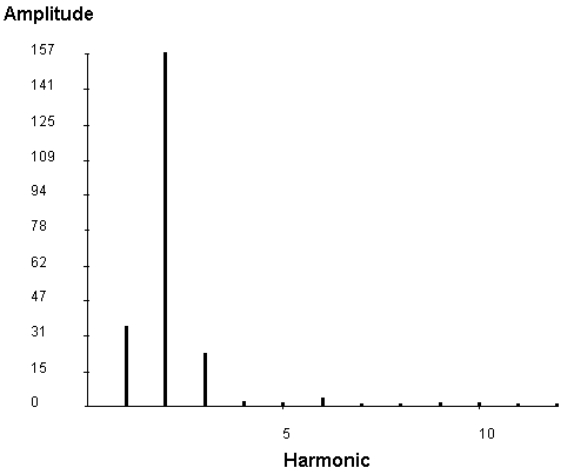


Figure 12. Average Harmonic Amplitudes of Filtered 1820 Hertz Call for Bird 2

After using the phase vocoder analysis to find the three average spectra and converting them to the Csound function tables, we concentrated on transcribing the wide variety of the other bird calls.

Transcribing the Bird Calls

This section describes transcribing the bird calls. As already shown, the spectra of these birds are fairly close to sine waves, so the most distinctive feature of each call is its expressive pitch contour. As shown in Figure 3, the four-note call used for Bird 1 has a flat frequency contour. The actual frequency, around 2000 Hertz, is quite close to B6, and it has a clear rhythm (see Figure 13). The four

sounds in this call use the amplitude envelope with a continuous rise and almost no decay shown in Figure 4. Although this motive seems quite simple, this little bird turned out to be very useful as a rhythmic marker in the composition described below.
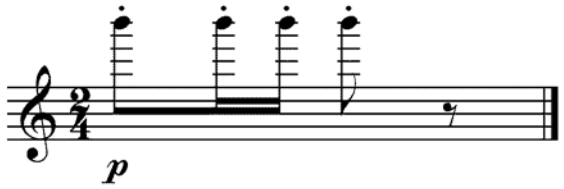


Figure 13. Transcription of Bird Call 1, Example 1.

We added a lower frequency for Bird Call 1, quite close to C#6 (around 1100 Hertz), using the first tone of a different bird's call (see Figure 14). That complete call has a rising motif, followed by an answering melody (from the same bird). The bird's rhythm is slightly different from Ayers' transcription, which adapts it to her musical purposes. Some of the bird tones have an amplitude modulation which she modelled using the previous flutter tongue design. (Ayers 2004)
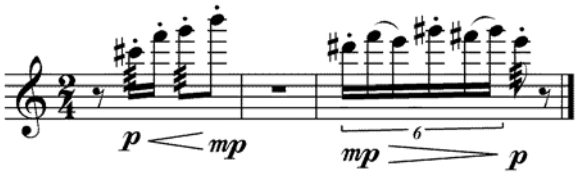


Figure 14. Transcription of Bird Call 1, Example 2.

The six-call motif used for Bird 2 (see Figure 15) has an overall flat contour, with preceding portamento grace notes providing expressive interest. The fundamental frequency of this call is around A6 at 1820 Hertz, with the grace note beginning at around 1650 Hertz (about G#6). Changing the grace notes to G natural made them fit better with the pentatonic E minor scale on the native American flute.
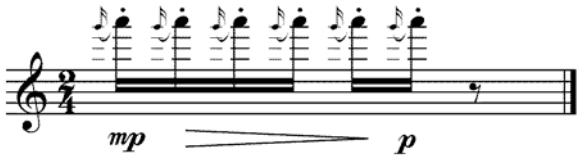


Figure 15. Transcription of Bird Call 2, Example 1.

A number of other interesting calls proved useful for the piece; Figure 16 illustrates a particularly striking pair of calls. Many times, a bird repeated its call with noticeable microtonal variations on the pitches, and it was quite fun to exploit this characteristic in the composition.
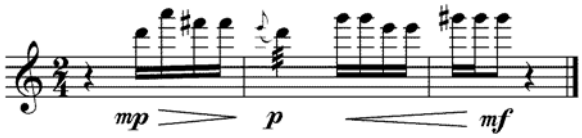


Figure 16. Bird Call Example 3.

Musical Use of the Design

From the "Dreamways" of the Iroquois to the "Dreamtime" or "Dreaming" of the indigenous Australians, ancient peoples have left their stories carved or painted on stone. In *Rock Art in the Dream World*, Ayers has created a musical story about these people, who travelled through their natural world and their dreams to create rock art. The live native American flute in E and *didgeridoo* in E represent two human characters. Accompanied by their synthesized dream selves, they stroll through a natural environment including a native American medicine rattle and recorded and synthesized Australian birds, crickets, kangaroos, rock taps and tap sticks.

The composition is divided into the following sections:

Journey to the Art Place
Creating the Art
Reflecting on the Art
Finishing the Art
Into the Dream World
Return

The composition uses a number of other bird call transcriptions using the three spectra previously described to model all the birds. For example, both of the bird calls illustrated in Bird Call 1, Example 2 (see Figure 14) are transformed in both the real and synthesized dream flute parts. In the slower section of "Into the Dream World," some of the motives, including this one, are stretched into "slow-motion" time (see Figure 17).
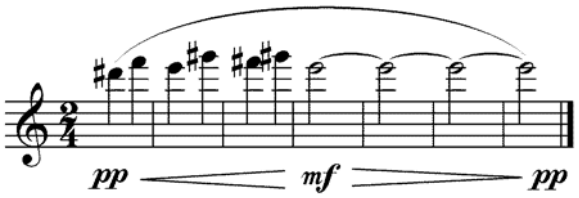


Figure 17. "Slow Motion" of Bird Call 1, Example 2.

The Flute Part

The native American flute part grew out of the motivic material transcribed from the bird calls. However, each flute typically has a range of slightly larger than an octave, with a basic pentatonic minor scale. Using two flutes an octave apart expanded the range, but the birds remain out of range of both flutes. Figure 18 shows parallel use of motive material from bird calls in the flute part. At times, the flute and birds play together, at other times they chase each other, with the birds keeping out of reach. The dream flute shadows ahead, behind or in harmony with the live performer, too.

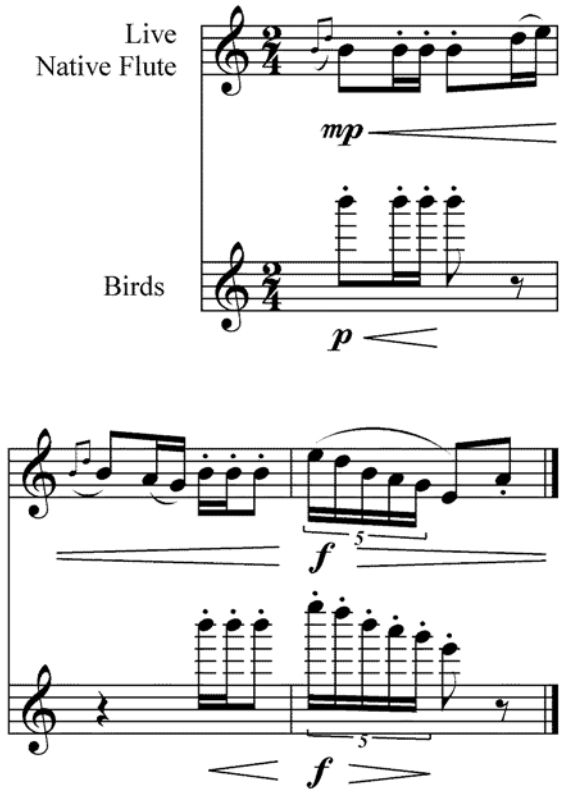


Figure 18. Parallel Motives in Flute and Bird Calls

The *Didgeridoo* Part

The live *didgeridoo* performer is free to improvise his part as he wishes. A demo version gives a sense of how his live performance may interact with his dream self. The synthesized *didgeridoo* used for the dream self and the demo live part is based on the design described in Ayers and Horner 2007.

Cricket Chirps

We were also able to synthesize a cricket chirp using the same design. The cricket makes longer single chirps and groups of shorter chirps. Each chirp is made up of a high frequency sound with amplitude modulation (see Figure 19).

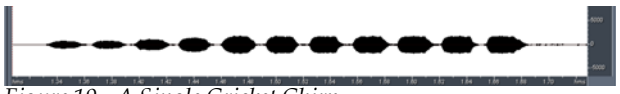


Figure 19. A Single Cricket Chirp.

The spectrum of the cricket chirp is somewhat similar to the spectrum of Bird Call 2, though for the cricket the second harmonic is the only one with significant amplitude (see Figure 20).

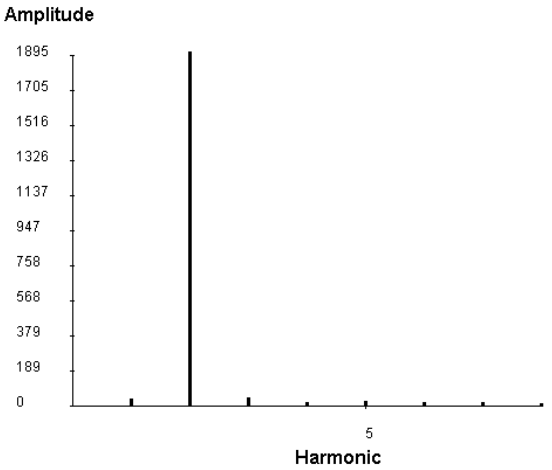


Figure 20. Peak Amplitude of Cricket Chirp.

Figure 21 shows several cricket motives, sounding an octave higher, that function as rhythmic markers in the composition.

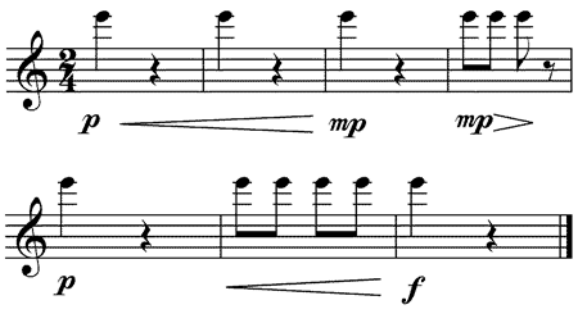


Figure 21. Cricket Chirps

Conclusion

We used additive synthesis to model the dynamic spectra and musical expression of some Australian birds. The Csound design captures the subtle characteristics of the bird calls, and also works for cricket chirps. In addition, Ayers transcribed the calls and used the design in the soundscape of a composition, *Rock Art in the Dream World*, for flute and *didgeridoo,* which demonstrates the expressiveness of the design.

Acknowledgements

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